

Computer Architectures

Exam of 07.02.2023 – part II

Question 1

Given two areas of memory, the first one containing byte constants and the second one being uninitialized, write the `copyData` subroutine in ARM assembly language, which copies the content of the first area of memory to the second one. The subroutine receives in input:

- the address of the first area of memory
- the address of the second area of memory
- the number of elements declared in the first area of memory

The procedure does not return any value.

The size of the second area of memory is higher than or equal to the size of the first area.

Example:

```
        AREA constants, DATA, READONLY
inputData DCB 3, -14, 15, -92, 65, 35, -89

        AREA variables, DATA, READWRITE
outputData space 12
```

Then, write the `insertionSort` subroutine, which receives in input:

- the address of an area of memory (READWRITE)
- the number of elements in the area (7 in the previous example)

The procedure does not return any value. It sorts the elements, rewriting the area of memory, by means of the insertion sort algorithm. The pseudocode of the insertion sort is the following (A is the array to sort):

```
1. i ← 1
2. while i < length(A)
3.     x ← A[i]
4.     j ← i - 1
5.     while j >= 0 and A[j] > x
6.         A[j+1] ← A[j]
7.         j ← j - 1
8.     end while
9.     A[j+1] ← x
10.    i ← i + 1
11. end while
```

In the example above, if the subroutine receives the address of `outputData`, at the end the area contains the values -92, -89, -14, 3, 15, 35, 65.

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Question 2

Extend the project developed in the previous question as follows.

Initialize timer 1 to count from 0 to 0xFF. When the timer counter reaches 0xFF, it is reset and it starts counting again; no interrupt is generated.

Declare an uninitialized array of char. The size of the array is MAX_VALUES, which is a positive constant that you can define as you prefer (e.g., MAX_VALUES = 20).

When button INT0 is pressed, you have to:

- Read the value of the timer counter of timer 1
- Assign the value to the first available element (i.e., not yet initialized) of the array of char. At the first pressure of the button, the value of the timer counter will be saved at position 0 of the array; at the second pressure it will be saved at position 1, etc. If all elements of the array have already been initialized, the value of the timer counter is not used.
- Alternatively switch on led 6 and led 7 to signal the correct read of the timer counter. At the first pressure of the button, led 6 is switched on and led 7 is switched off. At the second pressure, led 6 is switched off and led 7 is switched on. At the third pressure of the button, led 6 is switched on and led 7 is switched off, etc.

When button KEY1 is pressed, you have to:

- Switch off led 6 and 7
- Call the `insertionSort` subroutine, passing the address of the array of char and the number of initialized elements. The subroutine will sort the elements of the array, considering them as signed byte values.
- When the subroutine ends, switch on led 11.